Summary of Key Findings

This study evaluated various properties of sevoflurane as a new inhalational anesthetic. Sevoflurane was reported to have a boiling point of 58.5 degrees Celsius and a vapor pressure of 200 torr at 25 degrees Celsius, confirming volatility under normal conditions. The authors measured distribution coefficients at 37 degrees Celsius. The water coefficient is 0.36, corn oil coefficient of 41.6, and an olive oil coefficient of 53.4. This illustrates that sevoflurane's affinity for oil is higher than for water. However, liquid sevoflurane is not easily soluble in vegetable oils and can form two phases under some conditions, so it does not have unlimited miscibility. The study also found slight hydrolysis in water and slight reactivity with soda lime under closed-circuit anesthesia conditions.

Application to Design Project

The article is relevant since it provides evidence that sevoflurane has an affinity for oil, which can support investigating an oil based absorber. The large difference between the water and oil distribution coefficients makes me think a nonpolar liquid phase can capture sevoflurane more effectively than a typical water scrubber. However, this paper did not test a scrubber or palm oil, so it cannot necessarily verify palm oil's effectiveness as an absorber. Given that sevoflurane is not freely soluble in vegetable oils means that further investigation is required.

Specific Information for the Design project

- Sevoflurane boiling point: 58.5 degrees Celsius at 760 torr
- Vapor pressure: 160 torr at 20 degrees Celsius and 200 torr at 25 degrees Celsius
- Blood distribution coefficient at 37°C: **0.60 ± 0.07**.
- Water distribution coefficient at 37°C: **0.36 ± 0.01**.
- Corn oil distribution coefficient at 37°C: **41.6 ± 2.1**.
- Olive oil distribution coefficient at 37°C: **53.4 ± 1.2**.

3. Minimum Alveolar Concentrations and Oil/Gas Partition Coefficients of Four Anesthetic Isomers [3]

Source Authors, Journal, Date, Affiliation

D. D. Koblin, E. I. Eger II, B. H. Johnson, P. Collins, M. H. Harper, R. C. Terrell, and L. Speers. *Anesthesiology*, Volume 54, Issue 4, April 1981, pages 314–317. DOI: 10.1097/00000542-198104000-00011.

Summary of Key Findings

Four structural isomers, including isoflurane, were evaluated to compare potency to oil solubility. This study measured minimum alveolar concentration (MAC) in dogs and olive oil/gas partition coefficients at 37 degrees Celsius. For three of the four isomers, including isoflurane, the MAC values ranged from 1.4102.67 vol% at 1 atmosphere. The partition coefficients ranged from 90.8-96.6, indicating strong oil affinity. The fourth isomer with eh lower oil/gas partition coefficient had a much higher MAC, indicating a relationship between lipid solubility and potency.

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Application to Design Project

This article is relevant because it illustrates isoflurane has a strong affinity for oil. This further suggests that an oil-based absorber is something worth considering, instead of water-based scrubbing. The study spoke about olive oil specifically, not palm oil, which is more readily available, or long-term removal of isoflurane. As such, it supports further research into an oil-based absorber, but not that it would be an effective final design.

Specific Information for the Design project

- Evaluated isoflurane.
- MAC was measured in dogs.
- MACs ranged from 1.41-2.67 vol% at 1 atm.
- Olive/gas partition ranged from 90.8-96.6 at 37 degrees Celsius.
- The fourth isomer had an oil/gas partition coefficient of 25.8 and a MAC of 12.53%, showing the relationship between lipid solubility and potency.

Individual Design Updates

HOQ Table

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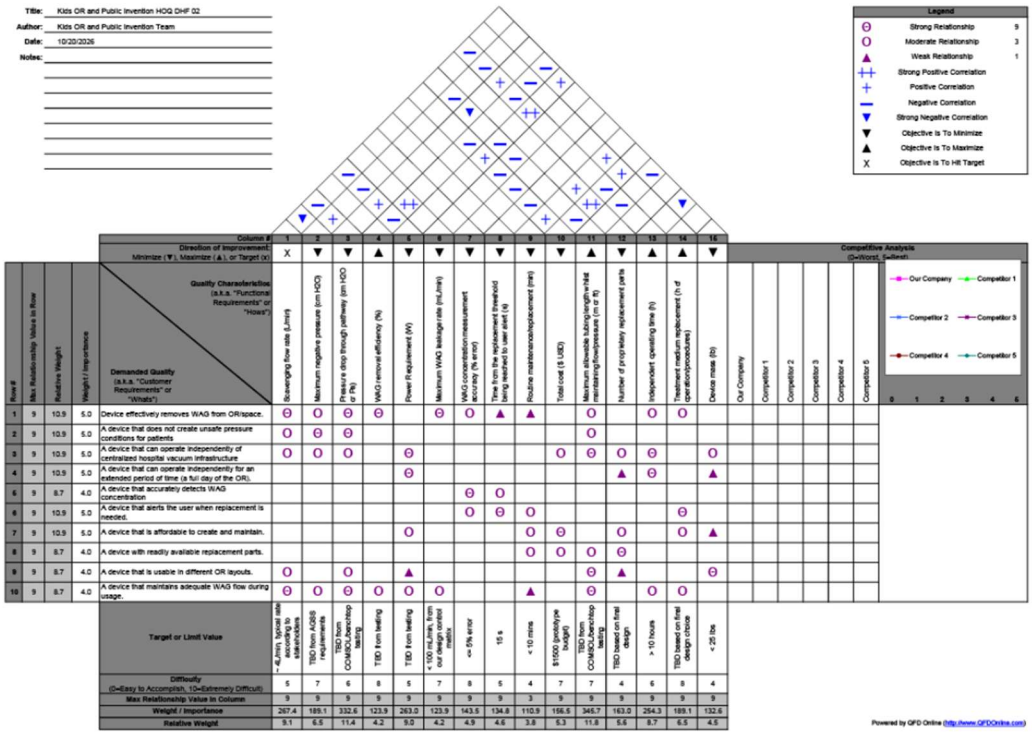


Figure 1. The team House of Quality developed for DHF 002. I created the entire House of Quality table, and user needs and design inputs were a mix of my own ideas as well as the rest of the team's.

Included in DHF 02. My discussion and inclusion here is about my contribution to the House of Quality and how my understanding of design requirements has changed.

I created the team's House of Quality (HOQ). The HOQ is meant to take the needs identified from stakeholders into measurable characteristics. This process was challenging because many of the known user needs were qualitative, while the HOQ relies on measurable and specific design inputs. There is much variability in OR layout and design as well as some missing information from stakeholders. The team is still creating the final concept, so many of these values are not known yet and will be discovered through modeling, testing, and additional stakeholder input.

One of the most important user needs is that the device effectively removes the waste anesthetic gas from the OR. This user need is part of multiple design inputs like scavenging flow rate, maximum leakage rate, and pressure drop in the scavenging pathway. The team has a stakeholder-provided flow value of around 4 L/min. Additional modeling and testing is necessary to determine the flow and pressure in the final system. Another important user need is that the system cannot create unsafe pressure conditions for the patient. This is particularly important as the team is considering using either a fan or blower. Excessive suction could occur if the active system is not far enough downstream from the breathing circuit. As such,

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maximum negative pressure and pressure drop are both important design inputs. The values are still TBD as the team does not yet have the information to establish limits. This is why I wanted my preliminary COMSOL modeling to take place as soon as possible, because these two primary variables will be a part of the model.

The HOQ helped me begin to think about maintenance, material availability, and tubing. Given the device is used in low-resource settings, affordability, availability of replacement parts, and ease of maintenance are crucial user needs. This influenced me to think of the creation of a refillable dry adsorption system, rather than a proprietary disposable cartridge/tray. It was difficult to determine tradeoffs for the design. Ultimately, we want to create a better solution for OR staff and patients, but we also want to consider the environmental cost. In remote areas, the release of untreated gases to the air can cause issues for limited and precious food and water sources.

Low Fidelity Drawing

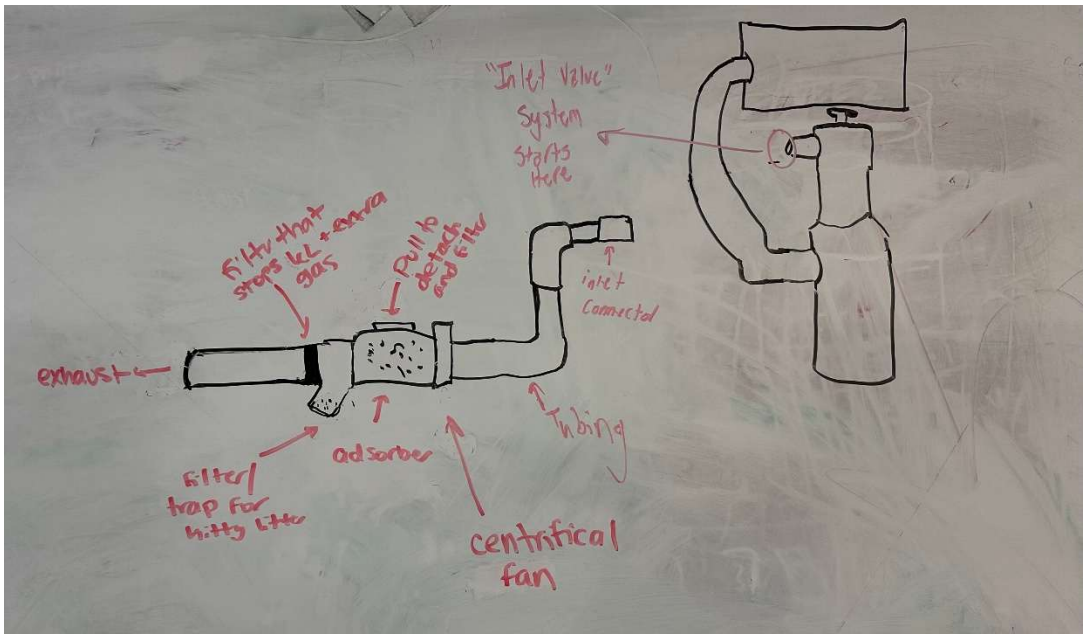


Figure 2. Team-developed low-fidelity drawing for a potential solution. This solution both scavenges and treats the waste anesthetic gases.

The low fidelity drawing was developed with all team members after revisiting the problem and identifying failing aspects of the current system. One of my main contributions was the idea of utilizing a great bend and treating or capturing the gas in the OR itself, rather than having to route it elsewhere. This would combat any of the current issues with the tubing like tripping hazards, tubing damage, and gas being routed to occupied areas of the hospital.

I also proposed using a centrifugal fan instead of an axial fan. Our pathway could include bends, treatment, and filters, all of which introduce resistance and pressure drop. This type of fan may be

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better for higher system resistance but could potentially be too strong for the patient's lungs if not sufficiently downstream. Another contribution I made is adding a final filter after the dry adsorbent section. This component would help keep any particles within the chamber and help minimize any unintended leakage. The exact material we have not decided yet and the pressure drop, retention, and adsorbent compatibility would need to be considered. While this preliminary design was not accepted by our advising team, it helped identify several items that should be considered during COMSOL and benchtop testing like pressure drop across tubing and treatment chamber, flow rate in the adsorbent bed, fan performance, and possible backflow.

COMSOL Modeling and Parameters

Parameter	**Why is this included?**
Tubing material/roughness	The main comparison between the current tubing and a smoother alternative like PVC. The roughness can cause pressure loss.
Tubing length	Longer tubing increases resistance and pressure drop, so good length whilst maintaining adequate flow is crucial.
Tubing diameter	Diameter affects flow resistance, so it should be controlled/compared if variable.
Number of bends	Bends create additional pressure losses and can reduce flow.
Pressure drop	A main output to compare the current tubing with the smoother tubing.
Outlet flow rate	This shows how much gas is moving through the system after accounting for tubing resistance and the fan/blower.
Fan/blower pressure flow	Determine if the fan is strong enough to overcome the resistance created by the tubing.

Figure 3. Preliminary COMSOL parameters to compare the current corrugated tubing with smoother alternatives as well as fan/blower effect on flow and pressure.

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My goal is to create a simplified model that focuses on the gas flow and pressure behavior first, before developing a more complex model. The reference model will be of the current passive system. While I could plan to vary tubing length and surface or roughness, I think it would be best to get an average from Courtney Ludick, KidsOR's lead biomedical engineer. I plan to compare corrugated tubing to smoother tubing like PVC or polyethylene.

For the proposed active design, I plan to create two separate models: one with a fan (axial) and another with a blower (centrifugal). This would help us evaluate which fan is better in our design in terms of performance. The primary outputs the team plans to evaluate are pressure drop, flow rate, and pressure distribution it the tubing pathway. These will help us determine if adequate waste anesthetic gas removal is occurring without unsafe suction conditions.

One challenge is that several design requirements are still being determined and finalized with stakeholders and literature review. Some of the requirements in the HOQ (Figure 2) will be found in the COMSOL modeling, like pressure drops and pressure conditions. As such, the initial COMSOL model will be used to compare configurations. The results will ultimately be compared to benchtop testing, allowing us to determine if the model predictions match real-world values.

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